MIL-C-60085D(MU)
21 September 1970
SUPERSEDING
MIL-C-60085C(MU)
14 April 1969

MILITARY SPECIFICATION

CABLE ASSEMBLY, FMU-7B/B

1. SCOPE

1.1 This specification covers one type of electric cable assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Film (Polyethylene Thin Gage).
QQ-S-781 - Steel Strapping, Flat.
TT-W-572 - Wood Preservative.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-B-117 - Bags, Sleeves and Tubing-Interior Packaging.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC 1325

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL14-15-972 - Cable Assembly, FMU-7B/B.
IDL14-15-972 - Cable Assembly, FMU-7B/B - Inspection Data List.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL14-15-972 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the cable assembly shall conform to the specifications and drawings listed on Drawing DL14-15-972 and subsidiary drawings.

3.2 Assembly. The cable assembly shall be assembled as shown on Drawing DL4-15-972.

3.3 Continuity. The electrical circuit of the cable assembly shall be continuous without exceeding a resistance of one ohm when tested in accordance with 4.4.4.1.

3.4 Open circuit. The cable assembly shall have an open electrical circuit between the adapter and each of the two terminals when tested in accordance with 4.4.4.2.

3.5 Leakage. The seals between the cable and fitting (Drawing C14-15-974), adapter (Drawing C14-15-973) and electrical connector (UG-260D/U) shall not leak nor shall there be any leakage through the cable insulation when the assembly is subjected to 35 + 5 pounds per square inch gage (psig) air pressure for 30 + 10 seconds while submerged in water in accordance with 4.4.4.3.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot of cables shall be produced in accordance with this specification for examination and test (see 4.3).

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3.7 Workmanship. The cable shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 200 cable assemblies shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination. Twenty-eight cable assemblies shall be taken at random.

4.3.2.2 For nondestructive tests. Twenty-eight cable assemblies (4.3.2.1) shall be taken for the leakage test. All cable assemblies of the preproduction lot shall be subjected to the continuity and the open-circuit tests.

4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample cable assemblies shall be examined in accordance with the classification of defects (4.4.3.2). The AQL's do not apply.

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4.3.3.2 For nondestructive tests.

(a) The sample cable assemblies shall be tested in accordance with 4.4.4.3.

(b) The preproduction lot of cable assemblies shall be tested in accordance with 4.4.4.1 and 4.4.4.2.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling. Sampling shall be conducted as follows:

(a) Classification of defects. In accordance with MIL-STD-105.

(b) Continuity. Each cable in the lot shall be taken for test.

(c) Open circuit. Each cable in the lot shall be taken for test.

(d) Leakage. In accordance with MIL-STD-105, level S-4.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. The sample cable assemblies and the level A preparation for delivery shall be examined and tested as follows:

(a) Classification of defects. In accordance with the classification of defects and MIL-STD-105.

(b) Continuity. In accordance with 4.4.4.1 and 4.4.5.

(c) Open circuit. In accordance with 4.4.4.2 and 4.4.5.

(d) Leakage. In accordance with 4.4.4.3, and MIL-STD-105, using an AQL of 1.0 percent defective.

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4.4.3.2 Classification of defects. (See inspection data list IDL14-15-972 for inspection equipment required to perform the examinations specified in this section.)

(a) Cable assembly, FMU-7B/B (Drawing D14-15-972).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Components missing or incorrectly assembled
102		Cable length incorrect (64 or 14-1/2 inches)
103		Pin (UG 260 D/U) not flush or below tolerance
104		Throat dimension (0.176) incorrect
105		Workmanship (3.7)

(b) Adapter (Drawing C14-15-973).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Thread (1-5/8-12UN-3A) incorrect
102		"O" ring groove and location incorrect
103		Eccentricity between throat dimension and external thread excessive

(c) Fitting (Drawing C14-15-974).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		"O" ring groove and location incorrect

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4.4.3.2 Classification of defects (continued).

(d) Clip, spring (Drawing D14-15-975).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Throat dimension (0.176) incorrect

(e) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Cables not properly cushioned
102	Component missing
103	Shipping containers incorrect
104	Marking incorrect or illegible

4.4.4 Tests. (See inspection data list IDL14-15-972 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Continuity test. An ohmmeter shall be connected to the center terminals of both plugs. The test shall be repeated across the outside terminals of both plugs.

4.4.4.2 Open circuit. An ohmmeter shall be connected to the center terminal and the outside terminal of one of the plugs. The meter shall read infinity on the RX1000 scale.

4.4.4.3 Leakage. The cable assembly shall be placed in a fixture, figure 1 or similar test apparatus, such that air pressure can be applied to the seals through the fittings, adapter and the cable between the fittings and adapter. The cable assembly shall be submerged in water containing a nonfoaming wetting agent at ambient temperature and the specified air pressure applied for the required period of time.

4.4.5 Acceptance/rejection criteria. If any cable assembly fails the tests in 4.4.4.1 and 4.4.4.2, that cable assembly shall be rejected. Items passing the above tests shall be returned to their lots.

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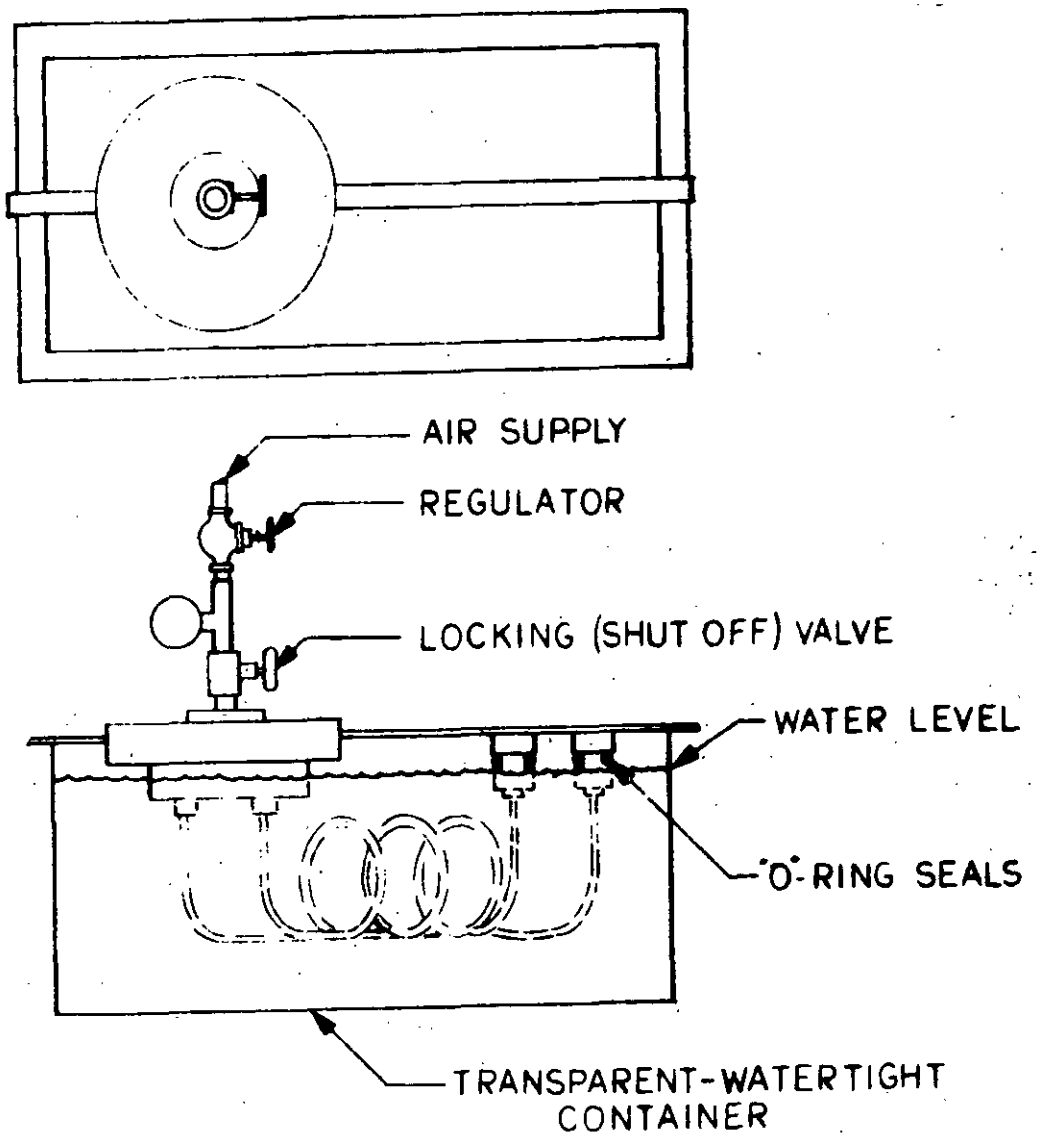


FIG 1 BUBBLE TEST FIXTURE

1. ATTACH CABLE ASSY TO FIXTURE & CLOSE VALVE
2. ADJUST PRESSURE REGULATOR TO 35 psi
3. IMMERSE CABLE ASSY.
4. OPEN VALVE
5. LOOK FOR BUBBLES

PART OF SPEC	CHEMICAL CORPS SPECIFICATION FIGURE
MIL-C-60085	DATE: 21 SEP 70 A136-301-61

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Level shall be as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 "O" ring. "O" rings required for one cable assembly shall be inclosed in a polyethylene bag conforming to L-P-378, having dimensions of approximately 2 by 3 inches. Bag closure shall be effected in a manner that will not damage the rings and will not allow spilling of contents.

5.1.1.2 Attaching hardware. Attaching hardware required for one cable assembly shall be inclosed in a polyethylene bag conforming to L-P-378, having dimensions of approximately 2 by 3 inches. Bag closure shall be effected in a manner that will not allow spilling of contents.

5.1.1.3 Cable assembly. Each cable assembly, with the required "O" rings and hardware, packaged in accordance with 5.1.1.1 and 5.1.1.2, shall be sealed in a bag conforming to type II, class E, style 1 of MIL-B-117, having dimensions of 9 by 11 inches. The cable assembly shall be compactly coiled in a manner that will not cause a permanent kink or set that would affect performance and enclosed in a wrap of kraft paper or other suitable noncorrosive cushioning wrap prior to sealing in the bag. Points or projections that may cause damage to the bag shall be padded or overwrapped. Included air volume shall be held to a practicable minimum when bag is sealed.

5.1.2 Level C. Each cable assembly, with the required "O" rings and hardware packaged in accordance with 5.1.1.1 and 5.1.1.2, shall be inclosed in a polyethylene bag conforming to L-P-378, minimum 4 mil thickness. Bag closure shall be effected in a manner that will not allow spilling of contents.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. Cable assemblies, packaged in accordance with 5.1.1, shall be packed 14 in a box conforming to class 2, style 4, for type 2 load of PPP-B-621, having inside dimensions of 27 by 9 by 10 inches, length, width and depth respectively. Material for the wood boxes shall be immersed in preservative conforming to composition optional of TT-W-572 for a minimum period of 3 minutes. Boxes shall not be loaded or marked for a minimum of 24 hours after preservative treatment. All interior faces of the boxes shall be lined with weather-resistant fiberboard conforming to PPP-F-320. Additional fiberboard pads shall be included as necessary to prevent free movement of the packages. Boxes shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to class B of QQ-S-781.

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5.2.2 Level C. The cable assemblies, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to class W5c or V3c of PPP-B-636. Quantity per box shall be 14 unless otherwise permitted by the procuring activity, and then the quantity shall not exceed the load limitations for the box. The contents shall fit the box snugly using fiberboard filler pads or other cushioning as required to prevent free movement of the packages. Boxes shall be closed in accordance with method II in the appendix of PPP-B-636.

5.3 Palletization. When specified (see 6.2), cable assemblies packed as specified, shall be palletized in accordance with Drawing D14-15-999.

5.4 Marking. In conjunction with information provided in the contract or order, marking shall be as specified herein.

5.4.1 Packaging. Each package (see 5.1.1.3 and 5.1.2) shall be marked with the Federal stock number and Department of Defense ammunition code number, nomenclature, quantity and unit of issue, level of packaging and month and year packaged, contract or purchase order number and lot number if applicable.

5.4.2 Packing. Boxes shall be marked in accordance with Drawing D14-15-986.

5.4.3 Pallet. Pallets shall be marked in accordance with Drawing D14-15-999.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required.
(c) Type of shipping container required.
(d) Palletization when required (mandatory for overseas shipment)
(e) Preproduction

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

(f) Pertinent identification marking.
(g) Applicable address marking.

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(h) When wood preservative is required for pallets (mandatory for overseas shipment).
(i) Shipment priority number, if applicable.
(j) Transportation control number, project code and required delivery date, if applicable.
(k) Any special marking required.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1325-A398

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Edgewood Arsenal
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Edgewood Arsenal, Maryland 21010

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